

The Rate Constant of Sequencing

On the Physical Identity of c

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Abstract

The constant c enters modern physics as "the speed of light in vacuum"—a description accurate in 1865 but progressively inaccurate ever since. By 1905, c governed the invariant structure of spacetime; by 1915, it regulated the coupling between geometry and energy-momentum; by 1983, it had become the definitional basis from which the unit of length is derived. Across this trajectory, light is a contingent occupant of the class of massless fields, all of which propagate at c for reasons unrelated to light in particular.

This paper identifies what c is across its full theoretical role. The argument proceeds from a feature acknowledged for nearly a century but not followed to its logical terminus: the time parameter t , across classical mechanics, quantum mechanics, and canonical quantum gravity, functions not as a dynamical variable but as an external ordering parameter. If t is an ordering parameter, the formalism requires a conversion constant specifying how much spatial extension corresponds to one unit of ordering. That ratio is c .

On this reading, the light postulate is a consequence rather than a postulate; the 1983 metrological decision institutionalises a conceptual order the physics had already reached; and the causal limit is not a dynamical barrier but a statement that no content of an ordered description can outrun the ordering that constitutes it. No equation is modified. The claim is interpretive: c is the rate constant at which temporal ordering is denominated in spatial extension.

Keywords: speed of light, problem of time, constants of nature, special relativity, Planck units, block universe, ordering parameter

1. What c Actually Does in Physics

In 1865, James Clerk Maxwell derived a quantity from the equations of electromagnetism. The permittivity and permeability of the vacuum— ϵ_0 and μ_0 —combined to yield a velocity: $(\epsilon_0\mu_0)^{-1/2}$. Its numerical value matched the measured speed of light. Maxwell drew the obvious conclusion: light is an electromagnetic wave, and its speed is fixed by the electromagnetic properties of the vacuum. The quantity was named accordingly. The name has persisted for a hundred and sixty years.

What the name has not kept pace with is the quantity's theoretical career.

In 1905, Einstein's special relativity elevated c from a measured speed of light to the invariant of the Lorentz group—the unique quantity on which all inertial observers agree, regardless of their relative motion or of whether light is present in the system being described. The mass-energy relation $E = mc^2$ expressed the conversion between two previously separate bookkeeping categories; the factor c squared plays no role involving the propagation of light and is not a speed in any operational sense. In Minkowski's 1908 reformulation of special relativity, the line element took the form

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

Here c appears multiplying the time coordinate, converting duration into a quantity commensurable with spatial extension so that time can participate in the geometry at all. The geometry is four-dimensional; its single currency is extension; time purchases admission to that geometry at the exchange rate c .

In general relativity, the Einstein field equations couple spacetime curvature to the energy-momentum tensor through a coupling constant that contains c to the fourth power. The electromagnetic constitutive relation $\epsilon_0\mu_0 = 1/c^2$ survives into the modern theory, but its role is now understood differently: it is not a fact about how fast light moves through the vacuum, but a statement about the structural ratio between electromagnetic and temporal units in the theory's natural denomination. In quantum mechanics, the Planck relation $E = h\nu$ couples energy to frequency; c converts between wavenumber and frequency in the relativistic dispersion relation $E^2 = (pc)^2 + (mc^2)^2$. In every relativistic quantum field theory, c bounds causal propagation as such—not the propagation of light in particular, but the propagation of any influence, luminous or not.

Since 1983, c has not been a measured quantity at all. The General Conference on Weights and Measures fixed its value at exactly 299 792 458 metres per second and redefined the metre as the distance light travels in vacuum in $1/299\,792\,458$ of a second. Length is now derived: a metre is a stated fraction of a second, converted at rate c . The hierarchy of the SI system now runs from the second—defined by the caesium hyperfine frequency—to the metre—defined via c —with the kilogram and other units following. The unit of spatial extension is derived from the unit of temporal duration through a conversion at rate c . This is the institutional commitment of metrology, and it has been in force for over forty years.

The theoretical record is unambiguous. c appears as a conversion factor between temporal and spatial quantities in the line element, as the invariant of the Lorentz group, as the exchange rate between mass and energy, as the coupling constant of electromagnetism and gravity, as the causal bound of relativistic field theory, and as the definitional basis of the metre. In none of these roles is c described accurately as the speed of a particular entity. Light happens to travel at c because light is a massless field, and massless fields propagate at the invariant speed of the Lorentz group, which is c —not because c is a fact about light. As Goldhaber and Nieto established comprehensively, if the photon were massive, light would travel at slightly less than c , while c itself—the invariant, the conversion factor, the coupling constant, the metrological basis—would remain exactly where it is, governing everything it governed before (Goldhaber and Nieto, 2010). The constant does not belong to light. Light belongs to the constant.

The question this inventory raises is not rhetorical. If c is the conversion factor between temporal and spatial quantities wherever they meet in physics, what is it the conversion factor between? Any ratio requires two commensurable kinds. If $c = (\text{extension})/(\text{duration})$, then the physical content of the ratio is: how much extension corresponds to one unit of duration. This is a meaningful question only if time and space are commensurable in some prior physical sense—only if there is something about the nature of the temporal parameter that makes its conversion into spatial extension determinate and exact. The answer to this question is already sitting in the foundations of physics. It has been sitting there since 1933.

2. The Status of t in Fundamental Physics

2.1 Classical and Quantum Mechanics: Time as External Index

The role of time in the mechanics of physics has a distinctive and persistent character that is easy to pass over and important not to. In classical Newtonian mechanics, the equations of motion govern the evolution of positions and momenta; t is the variable against which this evolution is tracked. It is the independent variable of dynamics. Nothing in the theory acts on t ; no equation governs it; it does not appear as a degree of freedom alongside positions and momenta. It is, in the language of modern physics, an external parameter—a label attached to states of the system, not a constituent of those states.

Quantum mechanics is, in almost every other respect, a revolution. Position becomes an operator, momentum becomes an operator, energy becomes an operator, angular momentum, spin, and the field amplitudes of quantum field theory all become operators. The mathematical transformation from classical to quantum physics is systematic and sweeping. Time alone passes through this revolution unchanged. The t in the Schrödinger equation

$$i\hbar \partial\Psi/\partial t = \hat{H}\Psi$$

is a classical c -number. It is not an operator. It has no spectrum. It does not participate in commutation relations. Pauli demonstrated in 1933 that time cannot, in principle, be promoted to a self-adjoint operator conjugate to a Hamiltonian bounded from below (Pauli, 1933)—

which means that no consistent quantum theory can treat time as an observable in the way it treats position or momentum. This is not a technical limitation to be overcome by future developments; it is a structural feature of what quantum mechanics is.

What is a parameter that labels states but is not itself a state? What does a quantity do when it indexes physical description without being an ingredient of what is described? It orders. The t of quantum mechanics does exactly what a sequence-label does: it distinguishes earlier from later, it allows the theory to say "the system was in state A before it was in state B," it provides the order in which the Schrödinger equation unfolds. It does not add anything to physical reality; it only orders what is there.

This is not a novel interpretation. It is the face-value reading of what the formalism does with t . Classical and quantum mechanics employ time as an ordering parameter—an external ordinal that labels physical states. That they do not say so explicitly, in these terms, does not affect the fact. The mathematical structure speaks clearly.

2.2 Canonical Quantum Gravity: Time Disappears

The external-parameter status of t becomes most consequential—and most diagnostic—when physicists attempt to apply quantum mechanics to the universe as a whole. In canonical quantum gravity, the quantisation of general relativity's Hamiltonian constraint produces the Wheeler-DeWitt equation:

$$\hat{H}|\Psi\rangle = 0$$

The wave function of the universe, $|\Psi\rangle$, satisfies a constraint whose formal content is that the total Hamiltonian annihilates it. The parameter t does not appear. There is no time derivative. In the canonical formulation, the wave function of the totality does not evolve (DeWitt, 1967).

This result is known as the "problem of time" (Isham, 1993; Kuchař, 2011; Anderson, 2012), and the physics literature has treated it primarily as a technical problem—an embarrassment to be engineered around. But the result deserves to be taken at face value before being circumvented, because at face value it says something that bears directly on the present argument: when physics describes a system with nothing external to it, the external parameter has nothing to attach to, and disappears.

The logic of the disappearance is transparent. An ordering parameter orders; it provides a "before" and "after" from which the perspective of an internal observer is built up. A system that contains everything contains no external vantage point; there is no observer outside the universe for whom the universe has a "before" and "after." The ordering parameter exists relative to a perspective, and where no such perspective exists, the parameter has no foothold. The Wheeler-DeWitt equation is the formalism registering, in its canonical structure, that t functions as a feature of description from within rather than as a dynamical ingredient of what is described. Whether this exhausts what time is, across all formulations of quantum gravity, remains contested; what is not contested is the formal role t plays in the canonical programme.

2.3 The Block Universe and the Convergence of Readings

The interpretation of time as a coordinate direction in a standing four-dimensional geometry—the "block universe"—has a distinguished lineage that begins with Minkowski's 1908 address and continues through Putnam (1967), Price (1996), and the contemporary philosophy of physics (Saunders, 2002). In the block universe reading, the four-dimensional manifold simply is; succession is how the structure is read from within, not how it comes to be. Past, present, and future are equally real directions in a geometry that does not itself flow. Einstein's late statement—that the separation between past, present, and future has, for those who understand physics, the character of an illusion, however stubborn—stands as the informal summary of this trajectory.

Gödel's 1949 contribution placed the point on rigorous mathematical ground (Gödel, 1949). Gödel's rotating universe solutions of the Einstein field equations admit closed timelike curves—loops in spacetime along which an observer could, in principle, return to their own past. In such a universe, there is no global foliation of spacetime into "earlier" and "later" slices. There is no consistent assignment of a universal time ordering. Gödel drew the moral explicitly: the objective lapse of time, the "becoming" that common experience presents as an inescapable feature of reality, is not guaranteed by the field equations of general relativity. It is not an intrinsic property of the relativistic physical world. It is a feature of certain solutions, not a feature of the theory.

Taken together, the three strata of fundamental physics converge on a single reading of t : an external ordering parameter, present where an internal descriptive perspective exists and absent where no such perspective is available, a feature of reading the physical structure rather than a constituent of the structure read. This reading is not this paper's proposal; it is this paper's inheritance—a summary of what the formalism of physics, examined at its foundations, already exhibits.

The inheritance generates a precise question. If t is an ordering parameter, then the ordering has a denomination. Something fixes the ratio between one unit of the ordering and the corresponding extension in the physical structure. A bare ordering is a list; physics is not a bare list—its time parameter couples quantitatively to every dynamical equation, every conservation law, every expression of causality. The denomination must be fixed. Where in the constant structure of physics is the ratio between sequence and extension already sitting?

2.4 Historical Convergence of the Structural Role of c

The reading of c as a conversion constant between temporal and spatial quantities is not, on examination, an abrupt departure from the trajectory of physics. It is the terminus of a line that several of the twentieth century's most careful thinkers followed to its edge without quite crossing.

Minkowski's 1908 address made the identification operationally. By writing the line element as $ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$, Minkowski did not merely combine space and time into a single manifold; he installed c as the exchange rate at which the temporal coordinate purchases admission to the spatial geometry. The ct that appears in every relativistic expression is not a notational convenience; it is the result of the conversion already performed. Minkowski's formalism treats the temporal coordinate as requiring denomination before it can enter the

geometry, and c is the denominator. He did not name this function of c , but the structure of his line element performs it exactly.

Eddington drew the implication further. In his analysis of the constants of nature, Eddington observed that c belongs to a different class from constants such as the electron charge or the gravitational coupling: it is not a property of a particular interaction but a structural ratio embedded in the metric itself (Eddington, 1920). A constant of interaction describes how entities of one kind respond to entities of another; a structural ratio describes the denomination of the manifold in which all interactions take place. Eddington did not have the language of ordering parameters; he had the observation that c behaves like a conversion, not like a coupling.

Weyl, in the same period, noted that the conformal structure of spacetime—the light-cone structure that determines causal order—is more fundamental than the metric itself, in the sense that the causal ordering of events is the primary physical datum from which the metric can be reconstructed, up to a conformal factor (Weyl, 1918). Malament later proved this rigorously: the causal structure of Minkowski spacetime determines its conformal geometry, fixing the metric up to an overall scaling factor (Malament, 1977). From the sequencing perspective, this result is significant: it establishes that the ordering of events is the prior physical datum, from which the spatial structure is recoverable. What the causal structure leaves undetermined—the physical conversion between temporal and spatial measurements required to assign dimensional units—is precisely what c supplies. The causal structure determines the conformal geometry; c fixes the dimensional conversion that completes the metric.

Reichenbach's analysis of the conventionality of simultaneity approached the same structure from the epistemological direction (Reichenbach, 1928). His ϵ -conventions for synchronisation show that what is genuinely objective in relativistic spacetime is not the assignment of simultaneous moments—which is partly conventional—but the causal order of events and the light-cone structure that enforces it. The conventional element is the choice of how to divide the interval between causal relations into "temporal" and "spatial" components; what is not conventional is the invariant quantity c that relates those components. Reichenbach's analysis identifies c as the fixed point around which the conventional choices are made, not as one of the conventional choices itself.

Taken together, these contributions form a recognisable trajectory: the identification of causal order as primary, the recognition of c as a structural constant of the manifold rather than a property of light, and the observation that temporal and spatial quantities require conversion before they can be related. What the present paper adds to this trajectory is the identification of the underlying reason: c converts temporal ordering into spatial extension because temporal ordering is what the time parameter is, and spatial extension is what it must be converted into before it can enter the geometry. The trajectory of twentieth-century physics points to this identification; the present paper names it.

3. The Conversion Constant

3.1 The Planck Architecture

The natural-unit system of physics—the system in which the constants c , G , and $\hbar$ are set to unity—expresses dimensional reduction in its most radical form: all physical quantities are expressed in a single unit, because the ratios between the fundamental dimensional quantities are fixed by the constants of nature. The Planck units are the quantities that result when this reduction is completed: the unique length, time, mass, and temperature constructible from c , G , and $\hbar$ alone. They are:

$$l_P = \sqrt{(\hbar G/c^3)} \approx 1.616 \times 10^{-35} \text{ m}$$

$$t_P = \sqrt{(\hbar G/c^5)} \approx 5.391 \times 10^{-44} \text{ s}$$

Their ratio is:

$$l_P / t_P = c$$

This identity is algebraically immediate and has been universally treated as such—an unavoidable consequence of the definitions, carrying no independent content. That treatment is mistaken, and the nature of the mistake is the subject of this section.

An identity between two quantities is silent about which is primitive. From $A = B/C$ one may consistently regard B as derived from A and C , or A as derived from B and C . The algebra licenses both readings. The choice between them is a physical and interpretive question, not a mathematical one. The standard reading of $l_P/t_P = c$ takes c as primitive under the description "the speed of light" and regards the Planck units as derived curiosities—scales at which quantum gravitational effects become important, attached to no established mechanism and possessing no operational significance at accessible energies. On this reading, the identity is trivial: a ratio between two quantities constructed from c returns c .

But the conclusion of Section 2 opens a different reading. If t is an ordering parameter—the external label under which physical structure is registered—then the physical structure has a natural denomination in temporal terms: the Planck time t_P is the unique temporal unit constructible from the constants that govern quantum gravitational physics, and it represents, in a natural sense, the finest temporal grain at which distinct physical states can be distinguished. Similarly, l_P is the unique length unit constructible from those constants. The ratio l_P/t_P then answers a specific physical question: how much spatial extension corresponds to one unit of the natural temporal ordering? The answer is c .

On this reading, c is not a speed that happens to equal l_P/t_P . It is the conversion constant between the natural unit of ordering and the natural unit of extension—the fixed ratio that specifies how much space one unit of temporal sequence is worth. The Planck units are not derived curiosities; they are the native denominations of the ordering and of physical extension, and c is what relates them. The identity is not trivial; it is a structural self-description of the formalism.

3.2 The Identity Is Not Circular: What Primitivity Assignment Does

At this point a precise objection must be met. The Planck units are defined from c (together with G and $\hbar$). Their ratio therefore returns c as an algebraic consequence of how they were constructed. To present $l_P/t_P = c$ as evidence for anything is to argue in a circle.

The objection is correct about the algebra. It misidentifies the subject of the argument.

The present claim is not that the identity $l_P/t_P = c$ constitutes independent evidence for the sequencing reading. The claim is that the identity is consistent with two different primitivity assignments—two different specifications of which quantities are fundamental and which are derived—and that the choice between them is not algebraically determined. One may take c as primitive and regard l_P and t_P as derived. Or one may take the natural units of extension and ordering as primitive and regard c as their ratio. The first reading is the standard one. The second reading is the one proposed here.

The two readings are algebraically identical. They differ in what they explain and what they assume. On the first reading, the conversion factor between time and space is a brute fact about the world, named for the phenomenon whose measurement first disclosed it. On the second reading, the conversion factor follows from the nature of the temporal parameter and the character of physical extension. The second reading does explanatory work that the first does not. Section 4 exhibits this work explicitly, by tracing five features of established physics that, on the first reading, must be assumed and, on the second reading, follow as consequences.

A reading is not circular because it employs the same mathematics as a competing reading. If it were, no two interpretations of the same formalism could ever be compared. Readings are evaluated by their explanatory reach, their internal consistency, and their capacity to convert assumptions into consequences. The sequencing reading is presented on those terms.

4. Five Consequences

4.1 The Light Postulate

Einstein's second postulate of special relativity—that the speed of light is the same in all inertial frames of reference, regardless of the motion of the source—is the most empirically successful unexplained assumption in the history of physics. A century of precision tests has confirmed it without exception. A century of theoretical development has not explained it. The postulate is assumed; the kinematics of spacetime is built upon it; and the question of why any quantity should be frame-invariant when all velocities are frame-dependent receives, within the standard reading, no answer.

The structure of the difficulty should be stated precisely. A velocity is the rate of change of position with respect to time, measured by a particular observer in a particular frame. Velocities transform under the Lorentz group; observers in relative motion assign different velocities to the same object. This is not a defect of the theory; it is its central result. The frame-dependence of velocities is what the Galilean and Lorentz transformations formalise. Given this, the existence of one velocity—just one—that all observers assign the same value is not a

generalisation from observed behaviour. It is an exception to the entire structure of how velocities work. The postulate asks us to accept this exception at the foundation of the theory without accounting for why the exception exists.

On the sequencing reading, there is no exception and no postulate requiring explanation. A velocity is a property of moving content—something whose spatial position changes with the temporal ordering. Properties of content are described from within frames, and therefore vary between them. The conversion constant between the ordering parameter and spatial extension is not a property of any content; it is a property of the parametrisation itself—the denominating structure within which all content and all frames are embedded. Every observer, in every frame, describes physical reality using the same temporal ordering; therefore every observer employs the same conversion constant between that ordering and spatial extension. The constant cannot vary between frames for the same reason that the exchange rate between the index and the indexed cannot vary: it is not inside the description, but beneath it.

What was called the light postulate is, on this account, not a postulate about light. It is a consequence of the nature of the ordering parameter. Observers measure the same c because they are all using c to convert their temporal indices into spatial extension, and the conversion rate is common to the formalism before any frame is selected. Light travels at this rate because light is massless, and massless fields propagate at the invariant speed—the speed set by the conversion constant of the ordering. The photon-mass literature confirms this reading by its counterfactual: if the photon acquired a small mass, it would propagate at slightly less than c , while c itself would remain the invariant of spacetime (Goldhaber and Nieto, 2010). Light can depart from c without affecting c . This is the signature of a constant that was never, fundamentally, a fact about light.

4.2 The Ubiquity of c

The appearances of c outside electromagnetism and optics are, on the standard reading, a series of coincidences requiring separate explanations. Why does the energy-mass conversion $E = mc^2$ involve c squared? Why does the vacuum constitutive relation tie the electromagnetic constants to c ? Why does the gravitational coupling constant in the Einstein field equations contain c to the fourth power? Why does c appear in the relativistic dispersion relation, in the causal structure of field theories, in the Lorentz factor that governs time dilation? Each of these appearances has a derivation that traces it back to the Lorentz invariance of the theory, but the question of why Lorentz invariance concentrates so much theoretical weight on this particular constant is not answered by pointing to the derivations. The ubiquity of c is a pattern, and patterns that appear across the full breadth of a theory invite a unified explanation.

The sequencing reading provides it. Every appearance of c in the inventory above is an appearance of the conversion constant between the ordering parameter and spatial extension. The mass-energy relation $E = mc^2$ expresses the conversion between energy—whose natural unit is the inverse of temporal duration, via the Planck relation $E = \hbar\omega$ —and mass, whose natural unit is expressed in terms of length and time. The factor c^2 is the conversion applied twice: once for each power of the temporal ordering that participates in the relation. The vacuum constitutive relation $\epsilon_0\mu_0 = 1/c^2$ ties the electromagnetic unit system to the ratio

between temporal and spatial units. The gravitational coupling constant $\kappa = 8\pi G/c^4$ converts between the geometric units of the curvature tensor and the mechanical units of the energy-momentum tensor—units that differ by the fourth power of the ordering-to-extension conversion.

Wherever a physical equation converts between quantities natively expressed in temporal terms and quantities natively expressed in spatial terms, the conversion rate c appears. This is not a coincidence; it is the structure of the formalism declaring itself. There is one conversion constant between the ordering parameter and physical extension, and every equation that crosses the boundary between temporal and spatial denomination must use it. c is ubiquitous because the ordering is ubiquitous—every dynamical equation tracks change against the ordering, and every spatial description must eventually be brought into relation with that ordering. The pattern is not mysterious once c is understood as the denomination factor of the ordering itself.

4.3 The 1983 Metrological Decision

The 1983 redefinition of the metre is, in the official presentation of the Bureau International des Poids et Mesures, a practical decision: the speed of light had become so precisely measurable that fixing its value and deriving the metre from the second offered a more stable realisation of the unit of length than any artefact or spectroscopic standard. The decision is presented as an improvement of convenience (BIPM, 2019).

The underlying logic deserves examination. For a quantity to serve as the fixed point from which the unit of spatial extension is derived, it must occupy a position in the definitional hierarchy that is prior to spatial extension itself. On the standard reading of c as a velocity—a ratio of distance to time—this priority is not straightforwardly available. A velocity, in its ordinary conception, presupposes the units of both length and time; it is a ratio between them, not a foundation for either. The 1983 decision does not, in its own terms, measure a velocity and then fix it: it fixes a numerical value for a physical constant and reconstitutes the unit of length from that value and the second. But the question of why this reconstitution is coherent—why c can occupy the role of conversion constant rather than derived ratio—is precisely what the standard reading leaves implicit. The sequencing reading makes the hierarchy explicit: the SI redefinition reveals an implicit structural priority of temporal standards over spatial ones, with c as the conversion relation between them.

On the sequencing reading, the decision is not only coherent but structurally motivated. c is the conversion constant between temporal ordering and spatial extension. The natural hierarchy therefore runs: temporal ordering is primary; spatial extension is derived from it via c . The second, defined by an atomic frequency, is the unit of temporal ordering. The metre, derived by multiplying a stated number of seconds by c , is the unit of spatial extension derived from the temporal unit at the conversion rate c . The SI system since 1983 is, in its metrology, a sequencing-reading system: it treats temporal ordering as primitive and spatial extension as derived. The practical motivation for the decision—the superior stability of the temporal standard—is a consequence of the same underlying structure.

The routine adoption of natural units, in which $c = 1$ and time is measured in metres, performs this derivation in calculational dress. Physicists who work in natural units directly measure time and space in the same unit, which is possible only because c converts between them. This practice is nearly universal in relativistic and high-energy physics. It is, in daily use, the sequencing reading applied as a working convention. The physics has operated on the sequencing reading for decades; only the semantics has not kept pace.

4.4 Time Dilation as Sequencing Allocation

Special relativity establishes, and experiment confirms without ambiguity, that a clock in motion with respect to an inertial frame accumulates less proper time than an identical clock at rest in that frame. The twin who travels returns younger. The Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$ quantifies the discrepancy. Muon lifetimes, GPS satellite corrections, and precision interferometry all confirm the prediction. The fact is not in question.

The standard presentation provides a correct derivation but an incomplete interpretation. The dilation follows from the geometry of the Minkowski metric; proper time along a worldline is the Minkowski length of that worldline, and a worldline that includes spatial displacement is longer in spatial terms but shorter in temporal terms than a worldline that does not. The geometry is correct. The persistent question—what, physically, happens to the moving clock?—receives the technically accurate but explanatorily opaque answer: nothing happens to it; the spacetime geometry differs.

The sequencing reading provides an ontological interpretation of the metric relation. The Minkowski line element, rearranged, reads:

$$c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

where $d\tau$ is proper time along the worldline. This can be written:

$$(c d\tau)^2 + (dx^2 + dy^2 + dz^2) = (c dt)^2$$

The geometric content of this relation is standard. What the sequencing reading adds is a way of reading what the relation says: $c dt$ is the total conversion of one unit of temporal ordering into spatial extension; this total divides, by the Pythagorean structure above, between advance through spatial extension (the displacement terms) and accumulation of proper interval ($c d\tau$). A system at rest converts its entire ordering unit into proper time: $d\tau = dt$. A system in motion distributes the conversion between spatial advance and proper time accumulation. The Lorentz factor is, on this reading, not a mysterious consequence of geometry but the division rule for a fixed total rate.

No new physics is introduced; no equation is modified; no prediction differs from the standard account. The four-dimensional velocity of every worldline has invariant magnitude c in both readings—this is the mathematical content of the Minkowski relation. What the sequencing reading offers is not an alternative derivation of time dilation but an ontological interpretation of why the invariant magnitude takes the value it does: c is the conversion constant of the ordering, and therefore sets the total rate that the metric relation partitions.

4.5 The Causal Limit

The upper bound on signal propagation velocity—the prohibition on influences travelling faster than c —is, on the standard reading, enforced dynamically. The relativistic momentum $p = \gamma mv$ diverges as $v \rightarrow c$ for any massive particle; reaching c would require infinite energy. This derivation is correct. It describes the steepness of the barrier without identifying what the barrier is or why it exists. The question "why is there a maximum speed?" is answered by pointing to the mathematical behaviour of the Lorentz factor, which is an account of the obstacle rather than an explanation of the limit.

On the sequencing reading, the limit is not a dynamical obstacle but a structural necessity. c is the rate at which the temporal ordering is denominated in spatial extension. Any physical influence is a transmission within the ordered description—a signal carried from one state to a later state as registered by the ordering parameter. For an influence to travel faster than c would be for a content of the description to outrun the ordering that constitutes the description. In some reference frames, such an influence would arrive before the ordering that defines "before" had been established at the destination. The relativity of simultaneity for spacelike separations, and the causal paradoxes that superluminal signalling produces, are the formalism registering exactly this impossibility.

A signal cannot outrun the ordering because the ordering is what makes "before" and "after" available for the signal to traverse. The causal limit is the ordering limit: nothing propagates faster than the rate at which the ordering unfolds, because that rate is the condition of there being a "before" and "after" for the propagation to navigate. The dynamical energy-divergence is the quantitative expression, within the formalism, of a logical impossibility: the impossibility of a content preceding its own ordering structure.

5. Consistency: The Lorentz-Invariance Constraints

Any reading that connects c to the temporal ordering at the Planck scale must address the stringent observational constraints on Lorentz invariance and, in particular, on any granularity of the spacetime manifold at that scale. Gamma-ray burst observations, particularly those of GRB 090510 by the Fermi-LAT collaboration, have placed bounds on energy-dependent photon velocity dispersion that rule out linear Planck-scale corrections to the dispersion relation at confidence levels exceeding several standard deviations (Abdo et al., 2009). More generally, the tests surveyed by Liberati (2013) and Amelino-Camelia (2013) constrain in-manifold Lorentz violation across many orders of magnitude.

The sequencing reading is untouched by these constraints. The constraints test for physical effects in the manifold—for actual dispersion of photons propagating through a spacetime that possesses residual granularity. The sequencing reading does not assert that the spacetime manifold is granular, or that Lorentz symmetry is violated at any scale, or that there are discrete gaps in the manifold between t and $t + t_P$. It asserts two things only, both interpretive: that t functions as an ordering parameter, and that c is the conversion constant between the natural unit of that ordering and the natural unit of physical extension. The continuum formalism, with its exact Lorentz invariance, is retained without modification. The Planck units enter not as the

scale of physical discreteness but as the natural denomination of the ordering—the scale at which one unit of sequence is, by the constant structure of physics, worth one unit of extension. No physical content of the manifold changes. No dispersion is predicted. The observational constraints on Lorentz violation confirm what the sequencing reading expects: that within the manifold, physics is exactly Lorentz-invariant.

The distinction from discrete-spacetime programmes is therefore exact, not merely approximate. Those programmes propose that the manifold itself has structure at the Planck scale—gaps, lattice points, or some form of physical granularity that would produce measurable dispersion. The sequencing reading proposes no such thing. It proposes that the parameter labelling the ordering has a natural denomination expressed in Planck units, and that c is the ratio between the natural units of the two quantities the parameter converts. The denomination is a statement about the semantics of the parameter; it is invisible to measurements performed within the manifold, for the same reason that the exchange rate between a currency and its denomination cannot be detected by performing arithmetic in that currency.

6. Objections

Is the sequencing reading merely the natural-unit convention restated?—Setting $c = 1$ in natural units is the sequencing reading applied calculationally, but a convention is not an interpretation. The natural-unit convention is presented as an arbitrary simplification: convenient, reversible, and noncommittal about what the constant is. The sequencing reading claims that the convention succeeds—that it simplifies correctly rather than accidentally—because it reflects the actual relationship between temporal ordering and spatial extension. The difference between a convention that works and an interpretation that says why it works is precisely the difference between using a map and knowing what territory it represents. The explanatory work of Section 4 is only available under the interpretation, not under the convention.

Does a conversion constant of the ordering not imply a preferred frame?—It does not, and the derivation of the light postulate in Section 4.1 shows why. A preferred frame would follow if the ordering were a physical process at a particular location in spacetime—a cosmic clock relative to which all motion is measured. The sequencing reading asserts the opposite: the ordering is the form of physical description from within, instantiated identically by every observer, prior to the selection of any frame. Every observer employs the same conversion constant because every observer's temporal parameter is the same ordering parameter; the rate at which it converts to extension cannot differ between observers for the same reason that a grammatical structure cannot differ between users of the same grammar. The constancy of c across frames is not consistent with the sequencing reading; it follows from it.

What of gravitational time dilation, where clocks at different potentials tick at different rates?—The locally measured value of c is the same everywhere in general relativity; the equivalence principle guarantees this. Gravitational time dilation, like kinematic time dilation, is an allocation effect: the budget identity of Section 4.4 is written in a curved metric, and the

allocation of the sequencing budget between proper time and the gravitational potential's geometry varies between worldlines while the conversion rate itself is unchanged. Coordinate speeds of light vary with coordinates, as all coordinate quantities do. The constant of the theory—the c of local light cones, the conversion factor in every local frame—is one number everywhere. General relativity generalises the geometry in which the allocation takes place; it does not alter the rate constant of the ordering.

Is the proposal testable?—As an interpretation of an existing formalism, the proposal makes no numerical prediction that the formalism does not already make; its accountability is of the kind appropriate to interpretive theses, and it accepts three explicit commitments. First, it is committed to the invariance of c : any evidence that the constant is frame-dependent, or varies with energy in the way dispersion-based Lorentz violation would require, would falsify the reading. Second, it is committed to the exact Lorentz invariance of in-manifold physics: any positive detection of Planck-scale dispersion of the kind that discrete-spacetime programmes predict would be inconsistent with the reading's claim that the Planck scale is a denomination of the ordering, not a scale of physical discreteness. Third, it is committed to the external-parameter status of t : if a future fundamental theory succeeds in restoring time as a dynamical observable—an operator with a spectrum, a genuine degree of freedom—the premise of Section 2 fails and the reading with it. A century of the problem of time, and the structure of the Wheeler-DeWitt equation, constitute the current score on that commitment.

7. Conclusion

The physical constant c entered physics in 1865 carrying a description—the speed of light in vacuum—that was accurate to its origin and has been progressively misleading ever since. By 1905 it governed the invariant structure of spacetime, by 1915 it regulated the coupling of geometry to matter, by 1983 it had become the definitional basis of the unit of length. Through all of these transitions, physics preserved the original name while the original entity—light—receded to the status of a convenient probe.

This paper has traced the consequence of two features of fundamental physics that have each been well understood separately and never fully combined. The first is the external-parameter status of t : the time parameter of classical mechanics, quantum mechanics, and canonical quantum gravity is an ordering index—a label that sequences physical states without being a constituent of those states, present when an internal perspective exists and absent, as the Wheeler-DeWitt equation shows, when the universe is described as a whole. The second is the structure of c in the Minkowski line element and across the full inventory of its theoretical appearances: a conversion factor between temporal and spatial quantities, present wherever the ordering parameter meets physical extension. The combination is direct: if t is an ordering parameter, then c is the conversion constant between the ordering and extension—the rate constant of sequencing, fixing how much spatial extension corresponds to one unit of the temporal index.

This identification converts five foundational assumptions into derivable consequences. The light postulate—that c is frame-invariant—follows from the fact that a conversion constant of the ordering is common to all frames before any frame is selected. The ubiquity of c across physics—in mass-energy conversion, electromagnetic constitutive relations, gravitational coupling, and relativistic dispersion—follows from the fact that every cross-denomination equation in physics crosses the same boundary between temporal and spatial quantities, at the same conversion rate. The 1983 metrological decision—to derive length from time via c —follows from the structural priority of temporal ordering over spatial extension in the sequencing reading, and renders that decision conceptually transparent rather than merely convenient. Time dilation follows as a budget allocation: the total sequencing rate per unit of ordering is c , this budget divides between temporal accumulation and spatial displacement, and the Lorentz factor is the division rule. The causal limit—the prohibition on signals exceeding c —follows from the impossibility of a content of the ordered description outrunning the ordering that constitutes it; there is no "faster than the ordering" available for any physical signal to occupy.

No equation in physics is modified. No experimental result is reinterpreted numerically. What changes is the physical reading of the formalism: the number 299 792 458 is not, primarily, a fact about how fast light travels. It is a fact about the temporal ordering—about how much spatial extension one unit of temporal sequence is worth. The description that has accompanied this constant for a hundred and sixty years is the name of the probe, not of the current. The current is time.

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